Persistent Visual Memory: Sustaining Perception for Deep Generation in LVLMs

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Research Questions

- How does visual signal dilution occur in autoregressive LVLMs as text sequence length increases?
- Can a parallel visual memory module sustain visual perception throughout extended generation without disrupting autoregressive reasoning?
- What is the optimal architectural design for maintaining persistent visual access in long-context multimodal generation?

Core Idea

LVLMs suffer from "visual signal dilution" where visual tokens at the sequence start get progressively drowned out by growing textual context in the attention mechanism's softmax normalization. As text length increases, the visual attention mass decays as O(t^-1), creating a structural bottleneck for long-form generation requiring sustained visual grounding. PVM solves this by adding a lightweight parallel branch alongside the FFN that performs cross-attention exclusively on original visual embeddings, creating an independent retrieval pathway that bypasses the diluted main attention stream while preserving reasoning capabilities.

Key Formula

Visual Attention Mass

$$\Omega_V(t) := \sum_{k \in V} \alpha_{t,k} = \frac{\sum_{k \in V} \exp(s_k(q_t))}{Z_V(q_t) + Z_T(q_t, t)} = \frac{Z_V(q_t)}{Z_V(q_t) + Z_T(q_t, t)}$$

Ω_V(t) is the aggregate probability weight allocated to visual tokens V at generation step t. Z_V is the unnormalized visual attention mass (bounded by fixed visual tokens M), Z_T is the textual attention mass (grows with sequence length t). This formulation captures how visual signal gets diluted as textual context expands.

Visual Signal Dilution Theorem

$$\Omega_V(t) \leq \frac{\beta}{\beta + \mu \cdot t} = O(t^{-1})$$

β = M·exp(s_max) is the upper bound on visual mass, μ is the lower bound on average textual attention mass. This proves visual attention decays inversely with sequence length t, establishing the theoretical foundation for why sustained visual memory is necessary in long-context generation.

Key Figure

$$\Omega_{\mathcal{V}}(t) \leq \frac{\beta}{\beta + \mu \cdot t} = \mathcal{O}(t^{-1}) \tag{2}$$

Proof. Substituting the visual upper bound $Z_{\mathcal{V}} \leq \beta$ and the textual lower bound $Z_{\mathcal{T}} \geq \mu t$ into Eq. 1 yields the governing inequality: $\Omega_{\mathcal{V}}(t) = \frac{Z_{\mathcal{V}}}{Z_{\mathcal{V}}+Z_{\mathcal{T}}} \leq \frac{\beta}{\beta+Z_{\mathcal{T}}} \leq \frac{\beta}{\beta+\mu\cdot t}$. As t grows, the linear term $\mu \cdot t$ in the denominator dominates the constant β, implying $\Omega_{\mathcal{V}}(t) = \mathcal{O}(t^{-1})$. □

3.1.2 Phase II: The High-Magnitude Saturation Trap

In practice, $Z_{\mathcal{T}}$ eventually saturates due to effective attention limits. However, we argue that this saturation does not rescue the visual signal; rather, it locks the model into a Low-Attention Equilibrium. Let W_{eff} denote the effective textual window size. Since the effective textual window is typically orders of magnitude larger than the visual context ($W_{eff} \gg M$), the aggregate textual mass stabilizes at a high plateau $Z_{\mathcal{T}}^{sat}$. Thus, the system converges to: $\lim_{t\to\infty} \Omega_{\mathcal{V}}(t) \approx \frac{\mathbb{E}[Z_{\mathcal{V}}]}{Z_{\mathcal{T}}^{sat}} \ll 1$.

Although the dilution halts, the system stabilizes with a heavily skewed distribution. The visual signal remains structurally outweighed by the overwhelming textual momentum, making the model more prone to errors when precise visual grounding is required.

3.2 Empirical Verification

To validate our theory, we analyze attention dynamics using Qwen3-VL-8B-Instruct [5] on a subset of the COCO 2017 [43] validation set. We designed a "Blind Painter" Stress Test (see Appendix J). Specifically, we prompt the model to reconstruct the image textually with extreme detail, as if instructing a painter who cannot see the original work. This setup necessitates active visual retrieval at every generation step rather than just the beginning, creating a rigorous testbed that isolates signal dilution from general language generation.

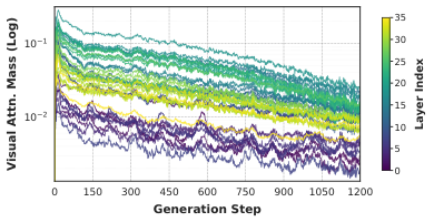

Figure 2: **Power-Law Decay of Visual Signal.** Log-scale analysis confirms that $\Omega_{\mathcal{V}}$ tightly follows the $\mathcal{O}(t^{-1})$ trajectory predicted by Theorem 3.1. This demonstrates that visual attention mass is structurally diluted, decaying inversely to the sequence length t.

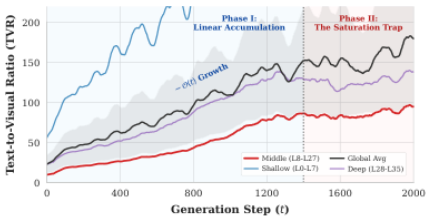

Figure 3: **Evolution of Textual Dominance.** The TVR trajectory effectively validates the proposed two-phase dilution mechanism. This reveals how initial probability competition saturates into a strict equilibrium, where textual priors structurally overwhelm visual signals.

Figure 2: Empirically validates the O(t^-1) power-law decay of visual attention mass across generation steps, confirming the theoretical dilution analysis and showing the phenomenon is most pronounced in intermediate layers (8-27) responsible for multimodal reasoning.

Limitations

- PVM relies on fixed local query assumption for theoretical guarantees, which may not hold during dynamic autoregressive generation
- Architecture requires manual selection of injection layers, though strided strategy (8,16,24) works best empirically
- Performance gains diminish with very large latent dimensions due to data-capacity mismatch in current training datasets
- Method tested primarily on Qwen3-VL series; generalization to other LVLM architectures needs validation